

One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

A stepwise model-selection study

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1 Introduction

This report studies one-day-ahead 1% Value-at-Risk (VaR) forecasting for QQQ daily log returns. Rather than listing models one by one, it follows a stepwise model-selection structure: start with simple volatility-based benchmarks, identify their weaknesses, and use those weaknesses to motivate the next improvement.

That structure matters because the main problem is not volatility forecasting alone. For 1% left-tail risk, the key issue is tail calibration. A model can track conditional variance reasonably well and still underestimate downside risk if its lower tail is too thin. The evidence in this study shows exactly that pattern. Gaussian baselines are too shallow, Student- t innovations help, asymmetric volatility helps somewhat, but the largest improvement comes from empirical-tail calibration through filtered historical simulation (FHS) and related bootstrap methods.

The model path is therefore:

- **RiskMetrics** ((= 0.94)) and **Gaussian GARCH(1,1)** as baseline volatility-VaR models,
- **Student- t GARCH(1,1)** once Gaussian tails prove inadequate,

- two extension paths: richer GARCH-family specifications and alternative nonparametric or direct-quantile approaches,
- and finally **empirical residual tail calibration**, which leads to the strongest results.

Across the common out-of-sample period of 848 trading days, the main recommendation is **GJR-FHS**, with **GARCH-FHS** a close second. **Historical Simulation** remains a serious benchmark. **GARCHX(VIX)** is retained as a comparison benchmark only. The simplified **CAViaR-SAV** result is interesting, but not convincing enough to overturn the final recommendation.

2 Historical Background

VaR became especially prominent in the 1990s as institutions sought a compact summary of downside market risk. J.P. Morgan’s *RiskMetrics* framework helped standardize daily variance-based VaR forecasting, which is why it remains a natural benchmark.

The econometric basis of modern VaR forecasting comes from conditional heteroskedasticity models. Engle’s ARCH model and Bollerslev’s GARCH extension formalized volatility clustering and provided a natural route from conditional variance forecasts to conditional quantile forecasts. Once a model for time-varying volatility is available, one-step-ahead VaR can be constructed from a conditional mean, a conditional standard deviation, and an assumed or estimated lower-tail quantile.

Evaluation then became as important as construction. Kupiec (1995) proposed unconditional coverage testing, while Christoffersen (1998) showed that correct hit frequency is not enough if violations cluster through time.

A second major development was the move away from purely Gaussian tails. Financial returns often exhibit heavier tails than Gaussian theory implies, which motivated Student-*t* innovations and semi-parametric methods. Barone-Adesi, Giannopoulos, and Vosper (1999) proposed Filtered Historical Simulation, which combines a volatility model with the empirical distribution of standardized residuals. That idea is central here because the strongest improvement in this report comes from empirical-tail calibration.

A different strand of the literature models the conditional quantile directly. The leading example is CAViaR (Engle and Manganelli, 2004). It is attractive because VaR itself is modeled directly, but in practice it requires careful optimization and adequacy testing.

3 Purpose Statement and Mathematical Development

3.1 VaR and the forecasting objective

Let r_t denote the QQQ daily log return,

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right).$$

For tail probability $\alpha = 0.01$, the one-day-ahead VaR is the conditional lower-tail quantile satisfying

$$\Pr(r_{t+1} < \text{VaR}_{t+1}(0.01) \mid \mathcal{F}_t) = 0.01.$$

For many models in this report,

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1}z_{t+1},$$

so the forecast takes the form

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{q}_\alpha.$$

This decomposition is useful because the models differ in two main ways:

1. how they forecast σ_{t+1} ,
2. how they obtain the tail quantile $\hat{q}_\alpha$.

The stepwise comparison in Section 5 is therefore a sequence of attempts to improve one or both parts of the VaR forecast.

3.2 Rolling out-of-sample design

The empirical study uses:

- 2,824 cleaned observations,
- a 70% initial training split,
- 848 one-step-ahead out-of-sample forecasts,
- an expected number of 1% violations equal to 8.48.

At each forecast origin, the model is re-estimated using information available up to that date and then used to forecast next-day VaR. This rolling design mirrors how VaR would be used in practice.

3.3 Violation indicator

Given a forecast $\widehat{\text{VaR}}_t$, define the hit indicator

$$I_t = \mathbf{1}\{r_t < \widehat{\text{VaR}}_t\}.$$

If the model is correctly calibrated at the 1% level, the sequence I_t should have mean close to 0.01 and should not exhibit systematic clustering.

4 Data Example

The study uses QQQ adjusted closing prices to construct daily log returns and uses VIX closing values only in the GARCHX benchmark. The purpose of this section is to make the model differences concrete through a single stressed example, not to rank models on the basis of one date.

A representative forecast date is 2025-04-04, when QQQ experienced a large negative return.

Date	Model	VaR	Return	Viol.
2025-04-04	GJR-FHS	-0.0796	-0.0641	No
2025-04-04	GARCH-FHS	-0.0706	-0.0641	No
2025-04-04	Student- t GARCH	-0.0631	-0.0641	Yes
2025-04-04	Historical Simulation	-0.0366	-0.0641	Yes
2025-04-04	RiskMetrics (0.94)	-0.0453	-0.0641	Yes
2025-04-04	GARCHX(VIX)	-0.0491	-0.0641	Yes
2025-04-04	CAViaR-SAV	-0.0629	-0.0641	Yes

The example shows the broad pattern seen in the full backtest. During stress, the empirical-tail models move substantially further into the left tail, while several baseline models remain too close to zero and are breached. The forecast and backtest plots used in the project support the same interpretation by showing how violations line up over time rather than only on a single date. Representative implementation snippets are collected in **Appendix A**.

5 Stepwise Model Development and Comparison

5.1 Baseline Gaussian volatility models

The natural starting point is the standard volatility-to-VaR pipeline used in many introductory treatments: forecast conditional variance, then map that forecast into VaR with a Gaussian quantile. The two baseline models are **RiskMetrics** ($(\lambda = 0.94)$) and **Gaussian GARCH(1,1)**.

RiskMetrics uses the EWMA recursion

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94,$$

with one-step forecast

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(\alpha).$$

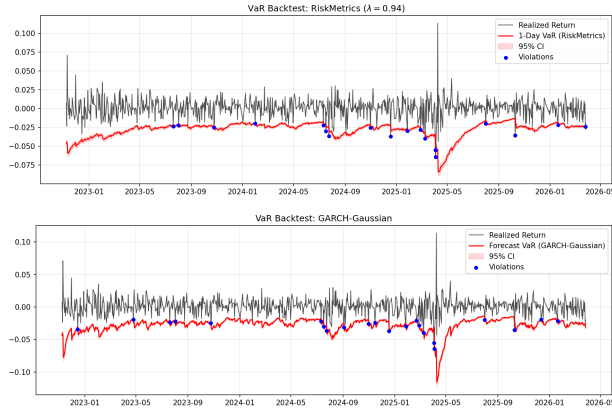
Gaussian GARCH(1,1) writes

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1),$$

with variance recursion

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
RiskMetrics ($\lambda = 0.94$)	18	2.12%	0.0043	0.3925	0.0117	0.0004458
GARCH- Gaussian	22	2.59%	0.0001	0.5970	0.0005	0.0004363



The paired baseline plots add something the table alone cannot: both VaR lines remain too close to zero during major stress episodes, especially relative to realized losses. The Gaussian GARCH line is a little smoother than RiskMetrics, but visually both models are too shallow in the left tail, which matches their excessive violation counts.

5.2 Heavier-tailed parametric improvement: Student- t GARCH

Once the Gaussian baselines underperform, the next step is to keep the GARCH volatility structure but allow heavier-tailed innovations. Student- t GARCH uses

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t,$$

where z_t follows a standardized Student- t distribution, while the variance recursion remains

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The VaR forecast is

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha^{(t)},$$

where $\hat{q}_\alpha^{(t)}$ is the fitted standardized Student- t 1% quantile.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GARCH(1,1) with Student- t innovations	13	1.53%	0.1480	0.1890	0.1482	0.0004131

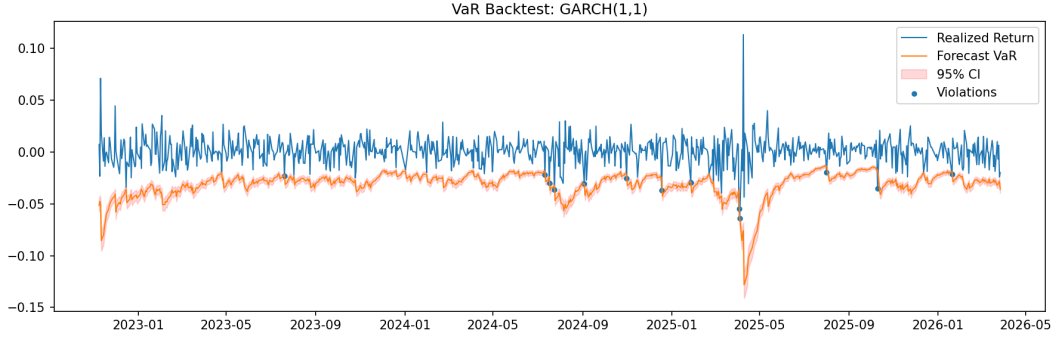


Figure 1: Student- t GARCH(1,1) backtest plot.

The Student- t GARCH plot shows the main visual gain from moving beyond Gaussian tails: the VaR line shifts deeper into the left tail during volatile periods. Even so, it is still breached too often in stress clusters, so the plot supports the same conclusion as the backtests: Student- t innovations help, but do not fully solve the calibration problem.

5.3 Extension path 1: GARCH-family extensions

Two richer volatility specifications are considered before moving to empirical tails.

5.3.1 GARCHX(VIX)

GARCHX(VIX) adds an exogenous volatility regressor to the variance equation,

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta x_t,$$

where x_t is standardized VIX. Economically, this is plausible because VIX summarizes market-implied uncertainty. Empirically, however, it is weaker than the leading models in this report.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GARCHX(VIX)	15	1.77%	0.0424	0.2621	0.0679	0.0004185

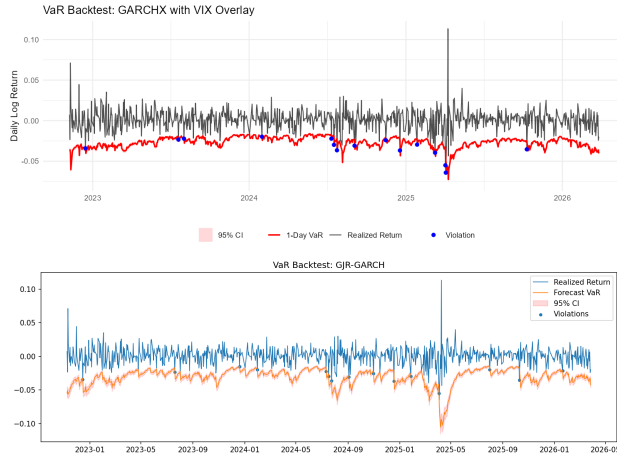
5.3.2 GJR-GARCH

A more promising extension is asymmetric volatility. GJR-GARCH uses

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2.$$

If $\gamma > 0$, negative shocks increase future volatility more strongly than positive shocks.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GJR-GARCH	15	1.77%	0.0424	0.4621	0.0972	0.0004069



The paired extension plots show that both models react coherently to changing volatility, but neither pushes the VaR line far enough into the left tail during stress. GARCHX(VIX) looks economically sensible, yet the final Gaussian tail mapping remains too weak. GJR-GARCH responds somewhat better to negative shocks, which is why it is the more promising extension, but asymmetry alone is still not enough.

5.4 Extension path 2: Historical / direct-quantile alternatives

5.4.1 Historical Simulation

Historical Simulation is a deliberately simple nonparametric benchmark. Using a rolling window of size $W = 250$,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \text{Quantile}_{\alpha}(r_{t-W+1}, \dots, r_t).$$

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
Historical Simulation	12	1.42%	0.2528	0.1569	0.1910	0.0004826

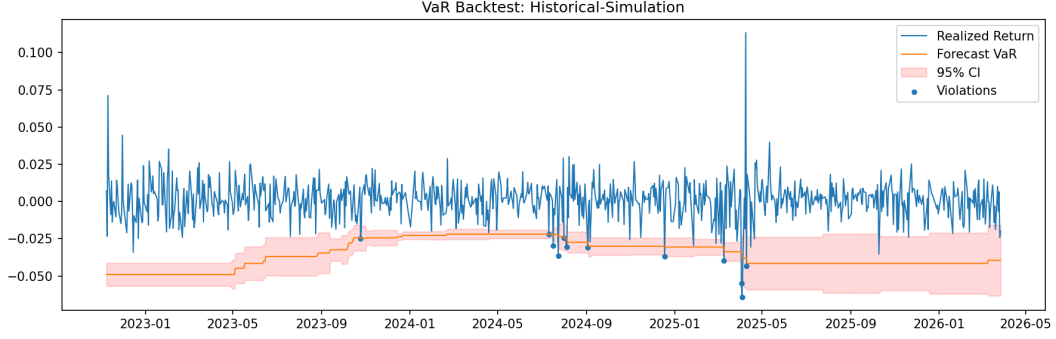


Figure 2: Historical Simulation backtest plot.

The Historical Simulation plot makes clear why this model remains a credible benchmark despite its simplicity. The VaR path adjusts in a visibly stepwise way and can lag after volatility regime changes, but it still tracks stressed periods well enough to remain competitive in the backtests.

5.4.2 CAViaR-SAV

CAViaR takes a different route by modeling the quantile directly. In the simplified SAV implementation used here,

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

and the left-tail quantile is recovered through

$$q_t = -v_t.$$

Parameters are estimated by minimizing the quantile check-loss objective,

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)), \quad \rho_\alpha(u) = u(\alpha - I(u < 0)).$$

The implementation is intentionally simplified: only the SAV form is included, optimization is relatively light, and the Dynamic Quantile adequacy test from the original CAViaR framework is not part of this study.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
CAViaR-SAV	6	0.71%	0.3664	0.0297	0.0625	0.0004010

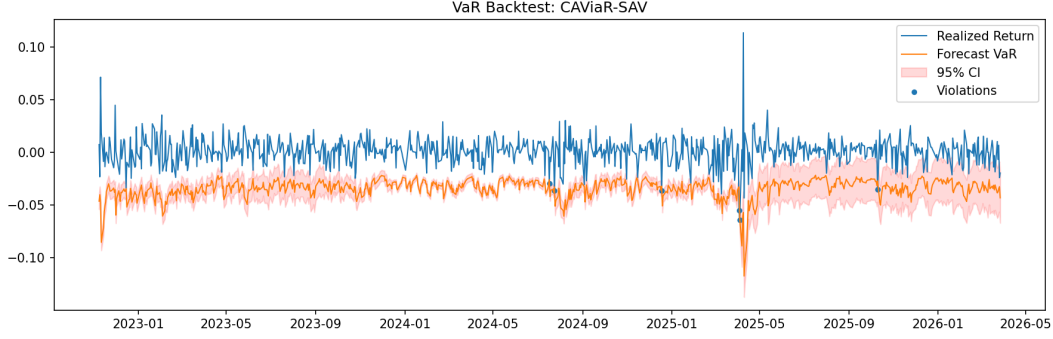


Figure 3: CAViaR-SAV backtest plot.

The CAViaR-SAV plot shows why the result is mixed rather than clearly successful. The quantile path often sits at a plausible level, which helps explain its good coverage and mean check loss, but the violation pattern is not evenly spread through time. The plot therefore supports the same point as the independence test: the level can look reasonable even when the timing of exceptions is not convincing.

5.5 Final improvement: empirical-tail FHS / bootstrap methods

By this stage the pattern is clear. Gaussian tails are too thin, Student- t helps, asymmetry helps somewhat, Historical Simulation is a serious benchmark, and CAViaR-SAV is intriguing but not fully convincing. The strongest remaining step is to keep the volatility model but estimate the lower tail empirically rather than through a fixed parametric law.

5.5.1 Stage 1: empirical tails with plain GARCH

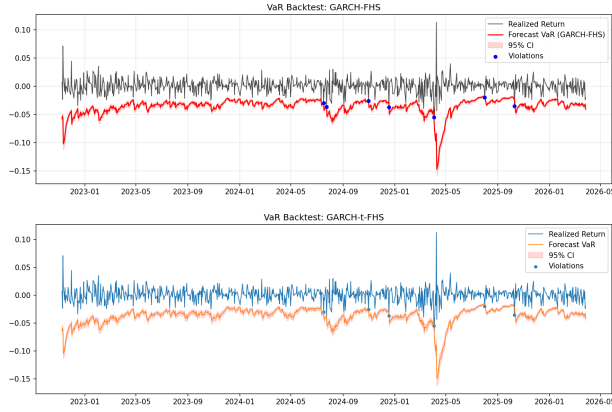
After fitting a volatility model, standardized residuals are computed as

$$\hat{z}_t = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t},$$

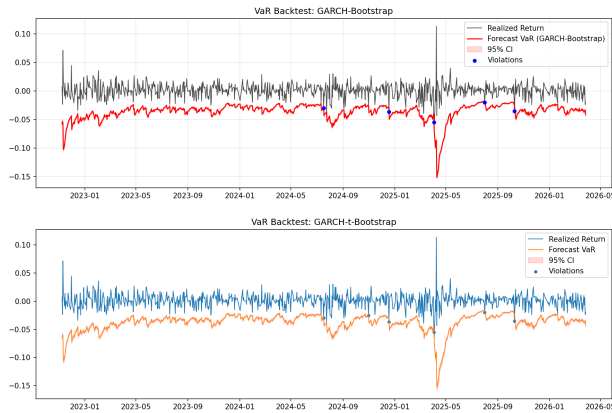
and the one-step forecast uses the empirical 1% quantile of those residuals,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}}.$$

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GARCH-FHS	7	0.83%	0.5984	0.7327	0.8212	0.0004126
GARCH- t -FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004258
GARCH-Bootstrap	5	0.59%	0.1934	0.8075	0.4166	0.0004154
GARCH- t -Bootstrap	6	0.71%	0.3664	0.7698	0.6371	0.0004302



The paired FHS plots show the main visual effect of empirical-tail calibration: both models move further into the left tail during stress than the purely parametric versions. The gain is not only in depth but also in timing, since the VaR line adapts more convincingly when volatility rises sharply.



The paired bootstrap plots look very close to the FHS plots, which is exactly what one would expect in one-step forecasting when both methods rely on empirical standardized residual tails. The bootstrap versions are slightly more conservative, but the main message is the same: empirical residual tails improve the forecast path materially relative to fixed parametric tails.

5.5.2 Stage 2: empirical tails with GJR-GARCH

The same empirical-tail idea can then be combined with asymmetric volatility:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2,$$

with the empirical residual quantile replacing the Gaussian or Student- t tail.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GJR-FHS	8	0.94%	0.8672	0.6961	0.9137	0.0004012
GJR- t -FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004171



The paired GJR empirical-tail plots complete the stepwise story. Both are visibly more responsive in the left tail than the earlier GJR-GARCH plot, but GJR-FHS gives the best overall balance between responsiveness and calibration: it deepens quickly in stress without drifting into excessive conservatism afterward. That visual balance is consistent with its strongest overall backtest performance.

5.5.3 Main lesson from the stepwise path

The main methodological lesson is that **empirical-tail calibration matters more than further refinements to the variance recursion alone**. Student- t helps. Asymmetry helps somewhat. But the decisive improvement arrives when the lower tail is estimated empirically.

6 Backtesting and Forecast Evaluation Metrics

The comparison uses four evaluation tools.

6.1 Kupiec unconditional coverage test

The null hypothesis is that the violation probability equals the target level α . If x is the number of violations in n forecasts and $\hat{\pi} = x/n$, the likelihood-ratio statistic is

$$LR_{uc} = -2 \log \left[\frac{(1 - \alpha)^{n-x} \alpha^x}{(1 - \hat{\pi})^{n-x} \hat{\pi}^x} \right].$$

A large p-value means the observed hit rate is plausible under the target 1% violation probability.

6.2 Christoffersen independence test

Correct hit frequency is not enough if violations cluster. The independence test compares an independent-hit model with a first-order Markov alternative. A low p-value indicates that violations are arriving in a dependent pattern.

6.3 Conditional coverage test

The conditional coverage test combines the previous two ideas:

$$LR_{cc} = LR_{uc} + LR_{ind}.$$

A good VaR model should perform reasonably well on both dimensions.

6.4 Check Loss

Because violation-based backtests ignore the size of forecast errors, the report also uses quantile check loss,

$$\rho_{\alpha}(u) = u(\alpha - I(u < 0)), \quad u_t = r_t - \widehat{\text{VaR}}_t.$$

Equivalently,

$$\text{CheckLoss}_t = (r_t - \widehat{\text{VaR}}_t) (\alpha - I(r_t < \widehat{\text{VaR}}_t)).$$

For non-violation days the penalty is small but nonzero, so overly conservative VaR still incurs loss. For violation days the penalty is much larger, especially when the realized loss falls far below the forecast threshold.

The model-level values reported throughout this report are **mean check losses**, i.e. the average of the 848 pointwise check-loss values for each model. This matches the project output file `model_check_loss_summary.csv`, where models are ranked by `MeanCheckLoss`.

7 Forecast Uncertainty and 95% Confidence Intervals

7.1 Why point VaR forecasts need uncertainty measures

A VaR forecast is an estimate rather than a fixed population quantity. It depends on estimated volatility, estimated or empirical tail quantiles, and finite-sample information. Reporting only a point forecast hides that uncertainty.

7.2 General quantile-CI logic

A common large-sample approximation for a quantile standard error is

$$\text{SE}(\hat{q}_\alpha) = \sqrt{\frac{\alpha(1-\alpha)}{nf(\hat{q}_\alpha)^2}},$$

which leads to the generic 95% interval

$$\hat{q}_\alpha \pm 1.96 \text{ SE}(\hat{q}_\alpha).$$

For location-scale VaR models,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha,$$

so quantile uncertainty translates directly into VaR uncertainty.

7.3 Model-specific CI constructions

The reported `lower_bound` and `upper_bound` values are model-specific rather than generated by one universal formula.

7.3.1 Gaussian parametric models

For Gaussian models such as RiskMetrics, Gaussian GARCH, and the package-based GARCHX benchmark, the Gaussian quantile standard error is mapped through

$$\hat{\mu} + \hat{\sigma} (\Phi^{-1}(\alpha) \pm 1.96 \text{ SE}(\hat{q}_\alpha)).$$

7.3.2 Student- t parametric models

For Student- t parametric models, the same logic is used with the standardized Student- t density and quantile in place of the Gaussian ones.

7.3.3 FHS and empirical residual quantile models

For FHS models, the empirical α -quantile of standardized residuals is first estimated and its sampling uncertainty is then mapped through the one-step volatility forecast.

7.3.4 Bootstrap models

For bootstrap models, standardized residuals are resampled, one-step returns are simulated, and the interval is built from the simulated predictive distribution.

7.3.5 Historical Simulation and CAViaR-SAV

Historical Simulation uses sample-quantile interval logic on recent raw returns. CAViaR-SAV does not produce a direct likelihood-based interval in the current implementation, so its interval is a proxy based on a recentered empirical return sample.

7.4 Interval-width comparison

Model	Mean CI width
CAViaR-SAV	0.02064
Historical Simulation	0.02064
GARCH- t -FHS	0.00783
GARCH-FHS	0.00688
GARCH(1,1) with Student- t innovations	0.00677
GJR-GARCH	0.00637
GJR- t -FHS	0.00586
GJR-FHS	0.00436
GARCH-Gaussian	0.00369
GARCH- t -Bootstrap	0.00368
RiskMetrics ($\alpha = 0.94$)	0.00368
GARCHX(VIX)	0.00353
GARCH-Bootstrap	0.00303

The leading models are not those with the narrowest intervals, but those that combine reasonable interval width with good calibration. Narrow intervals alone are not evidence of better risk forecasting if the VaR level itself is too shallow.

8 Integrated Final Comparison Table

Table 1 summarizes the full model comparison. Short labels are used only here to keep the table compact. They are defined in the note below.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss	Verdict
GJR-FHS	8	0.94%	0.8672	0.6961	0.9137	0.0004012	Best
GARCH-FHS	7	0.83%	0.5984	0.7327	0.8212	0.0004126	Strong
GARCH-t-FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004258	Strong
GJR-t-FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004171	Strong
GARCH-t-Boot	6	0.71%	0.3664	0.7698	0.6371	0.0004302	Strong
GARCH-Boot	5	0.59%	0.1934	0.8075	0.4166	0.0004154	Strong
HS	12	1.42%	0.2528	0.1569	0.1910	0.0004826	Benchmark
GARCH-t	13	1.53%	0.1480	0.1890	0.1482	0.0004131	Mid
CAViaR-SAV	6	0.71%	0.3664	0.0297	0.0625	0.0004010	Mixed
GJR-GARCH	15	1.77%	0.0424	0.4621	0.0972	0.0004069	Mixed
GARCHX(VIX)	15	1.77%	0.0424	0.2621	0.0679	0.0004185	Benchmark
RM(0.94)	18	2.12%	0.0043	0.3925	0.0117	0.0004458	Weak
GARCH-N	22	2.59%	0.0001	0.5970	0.0005	0.0004363	Weak

Note: HS = Historical Simulation; RM(0.94) = RiskMetrics with ($\alpha = 0.94$); GARCH-N = Gaussian GARCH(1,1); GARCH-t = GARCH(1,1) with Student- t innovations; GARCH-t-FHS and GJR-t-FHS use Student- t fitted volatility with empirical residual tails.

The table reinforces the main narrative. **GJR-FHS** is the strongest overall model because its violation rate is closest to the nominal 1% target and its backtest p-values are uniformly strong. **GARCH-FHS** is a close second and is the simplest strong empirical-tail recommendation. **Historical Simulation** remains a serious benchmark. **GARCHX(VIX)** is informative as a comparison benchmark but not competitive enough for the final recommendation. **CAViaR-SAV** remains a mixed result: attractive on mean check loss and unconditional coverage, but weakened by dependence in the timing of violations and by the simplified implementation.

9 Discussion

The empirical comparison points to a clear methodological conclusion. For one-day-ahead 1% VaR on QQQ, the hardest part is not volatility modeling by itself, but calibrating the lower tail once volatility has been forecast.

The Gaussian baselines fail first, showing that volatility dynamics alone do not solve the problem. Student- t innovations improve performance, which confirms that tail thickness matters. Richer

variance dynamics by themselves help less: GARCHX(VIX) is not persuasive in this setting, and GJR-GARCH helps only modestly until empirical tails are added. Historical Simulation performs well enough to remain a serious benchmark, while the simplified CAViaR-SAV result is interesting but not convincing enough to displace the leading models.

The decisive improvement occurs when standardized residual tails are estimated empirically. That is why the final recommendation is **GJR-FHS**, with **GARCH-FHS** a close second.

10 Limitations

Several limitations should be stated directly.

1. **Integrated comparison from multiple model runs.** The final comparison is coherent because all included models share the same 848-day out-of-sample horizon, but the full set of results is assembled from more than one run.
2. **GARCHX(VIX) remains a benchmark only.** The implementation is useful for comparison, but its final VaR mapping is still Gaussian.
3. **The CAViaR-SAV implementation is simplified.** Only the SAV specification is used, optimization is relatively light, and the Dynamic Quantile test is not included.
4. **Bootstrap forecasts are reproducible by construction.** The implementation resets the random seed inside the bootstrap forecast function, which improves reproducibility but makes the rolling bootstrap less genuinely stochastic than a fresh-resampling design would be.
5. **The conclusions are application-specific.** The ranking is for QQQ, a one-day horizon, and a 1% VaR target.

11 Conclusion

This report reorganizes the VaR study as a stepwise model-selection exercise. The starting point was a pair of Gaussian volatility-based VaR methods: **RiskMetrics** ($\alpha = 0.94$) and **Gaussian GARCH(1,1)**. These were natural baselines, but both produced too many violations, showing that Gaussian lower tails are too thin for QQQ one-day 1% VaR forecasting.

The next step was **GARCH(1,1) with Student- t innovations**. This improved performance, confirming that heavier tails matter, but it did not fully solve the problem. The extension path then split in two. In the richer-volatility direction, **GARCHX(VIX)** was not strong enough to justify itself as the main path, while **GJR-GARCH** suggested that asymmetry was somewhat useful. In the alternative-method direction, **Historical Simulation** proved to be a serious benchmark, and the simplified **CAViaR-SAV** implementation remained interesting but not convincing enough because of dependence failure and implementation limits.

The final and most important improvement came from empirical-tail calibration. Applying **FHS and related bootstrap methods** to plain GARCH already improved performance materially. Applying the same idea to **GJR-GARCH** improved further and produced the best overall final model. The final recommendation is therefore **GJR-FHS**, with **GARCH-FHS** a close second.

The central methodological lesson is straightforward: in this application, the largest improvement comes from **empirical-tail calibration**. That is why the report ends with **GJR-FHS** as the preferred model for one-day-ahead 1% VaR forecasting on QQQ daily log returns.

12 References

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13 Appendix A. Representative Implementation Snippets

This appendix includes short method-relevant snippets only. File loading and other boilerplate are omitted.

13.1 RiskMetrics recursion and VaR mapping

```
sigma2 = np.var(r[:init_window])
for i in range(init_window, n):
    sigma2 = lambd * sigma2 + (1.0 - lambd) * (r[i] ** 2)

sigma_next = np.sqrt(max(sigma2, 1e-12))
mu = float(np.mean(r))
var = mu + sigma_next * norm.ppf(alpha)
```

13.2 GARCH one-step VaR with Student-*t* innovations

```
y = returns * 100.0
model = arch_model(y, mean="Constant", vol="GARCH", p=1, q=1, dist="StudentsT")
res = model.fit(disp="off")

mu = float(res.params.get("mu", 0.0)) / 100.0
sigma2 = float(res.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / (100.0**2)
sigma = float(np.sqrt(max(sigma2, 1e-12)))
nu = float(res.params["nu"])
q = float(t.ppf(alpha, df=max(nu, 2.2))) * np.sqrt((nu - 2.0) / nu)
var = mu + sigma * q
```

13.3 GARCHX one-step forecast logic

```
y_adj <- r_train - mu
fit_garchx <- garchx(y_adj, order = c(1, 1), xreg = vix_train)
vix_T <- matrix(tail(vix_train, 1), ncol = 1)
pred_garchx <- predict(fit_garchx, n.ahead = 1, newxreg = vix_T)
sigma_next_x <- sqrt(as.numeric(pred_garchx))
var_garchx <- (mu + sigma_next_x * qnorm(alpha)) / 100
```

13.4 GJR-GARCH with empirical-tail mapping

```
y = returns * 100.0
model = arch_model(y, mean="Constant", vol="GARCH", p=1, o=1, q=1, dist="normal")
res = model.fit(disp="off")

mu = float(res.params.get("mu", 0.0)) / 100.0
sigma2 = float(res.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / (100.0**2)
sigma_next = float(np.sqrt(max(sigma2, 1e-12)))

std_resid = res.resid / res.conditional_volatility
std_resid = std_resid[~np.isnan(std_resid)]
q_emp = float(np.quantile(std_resid, alpha))
var = mu + sigma_next * q_emp
```

13.5 Historical Simulation

```
tail_sample = np.asarray(returns[-250:], dtype=float)
var = float(np.quantile(tail_sample, alpha))
```

13.6 CAViaR-SAV recursion and check-loss objective

```
@njit(cache=True)
def _caviar_sav_loss(params, r_scaled, alpha):
    b0, b1, b2 = params[0], params[1], params[2]
    v_prev = max(1e-6, -np.quantile(r_scaled, alpha))
    loss = 0.0

    for t in range(1, len(r_scaled)):
        v_t = b0 + b1 * v_prev + b2 * abs(r_scaled[t - 1])
        q_t = -v_t
        u = r_scaled[t] - q_t
        loss += u * (alpha - (1.0 if u < 0.0 else 0.0))
        v_prev = v_t
    return loss
```

13.7 FHS empirical residual quantile construction

```
std_resid = res.resid / res.conditional_volatility
std_resid = std_resid[~np.isnan(std_resid)]

q_emp = float(np.quantile(std_resid, alpha))
var = mu + sigma_next * q_emp
```

13.8 Bootstrap one-step simulation

```
std_resid = res.resid / res.conditional_volatility
std_resid = std_resid[~np.isnan(std_resid)]

np.random.seed(42)
boot_z = np.random.choice(std_resid, size=n_boot, replace=True)
boot_returns = mu + sigma_next * boot_z
var = float(np.quantile(boot_returns, alpha))
```

13.9 Check-loss calculation

```
u = y - q
check_loss = u * (alpha - (u < 0).astype(float))
mean_check_loss = float(np.mean(check_loss))
```

13.10 Backtesting logic

```
violation = int(realized < var)
lr_uc, p_uc = kupiec_test(v, alpha)
lr_ind, p_ind = christoffersen_independence_test(v)
lr_cc = lr_uc + lr_ind
```